THE FOURIER ENTROPY–INFLUENCE INEQUALITY FOR SYMMETRIC BOOLEAN FUNCTIONS

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ABSTRACT. We prove that every symmetric Boolean function f satisfies $\mathsf{H}[f] \le c\,\mathsf{I}[f]$ with $c = 931487/200000 = 4.657435$, improving the constant 12.04 of O'Donnell, Wright and Zhou. We bound the entropy lost under a random coordinate restriction and average the bounds over the successive dimensions. The sharp one-step constant is $K = 12 - \frac{9}{2}\log_2 3 = 4.8677\ldots$, attained by OR on three variables; retaining the dimension in the restriction argument gives a smaller entropy–influence constant. A noise-stability estimate for symmetric derivatives reduces the finite range to scalar inequalities. We certify them by outward-rounded interval arithmetic, use an exhaustive block-restriction census through dimension 20, and close the infinite range with a decreasing analytic bound. A sharper estimate for the shift of the entropy kernel improves the finite certificates. Without computer verification, the same noise-stability estimate gives $\mathsf{H}[f] \le 6.191\,\mathsf{I}[f]$. AND and OR show that the optimal constant is at least 4.

1. INTRODUCTION

For $f : \{-1,1\}^n \to \{-1,1\}$ write $f = \sum_{S\subseteq[n]} \widehat{f}(S)\chi_S$ with $\chi_S(x) = \prod_{i\in S} x_i$. By Parseval the squared coefficients form a probability distribution. Its entropy and the total influence are

$$\mathsf{H}[f] = \sum_S \widehat{f}(S)^2 \log_2 \frac{1}{\widehat{f}(S)^2}, \qquad \mathsf{I}[f] = \sum_S |S|\,\widehat{f}(S)^2 = \sum_{i=1}^n \mathbb{P}_x\big[f(x) \ne f(x^{\oplus i})\big],$$

where $x^{\oplus i}$ is x with its i-th coordinate flipped. The Fourier entropy–influence conjecture of Friedgut and Kalai [4] asserts that $\mathsf{H}[f] \le C\,\mathsf{I}[f]$ for an absolute constant C. It remains open. It is known for several classes of functions, among them symmetric functions [7, 5, 3, 1, 8].

A function is *symmetric* if $f(x)$ depends only on the Hamming weight $|x| = \#\{i : x_i = -1\}$. O'Donnell, Wright and Zhou [7] proved the conjecture for symmetric functions with constant 12.04.

Theorem A. *Every symmetric Boolean function f satisfies*

$$\mathsf{H}[f] \le c\,\mathsf{I}[f], \qquad c = \frac{931487}{200000} = 4.657435 < 4.6575.$$

For a coordinate i let $f_{i\to\pm1}$ be the two restrictions, $\mathsf{I}_i[f]$ the influence of i, and

$$\Delta_i[f] = \mathsf{H}[f] - \tfrac{1}{2}\big(\mathsf{H}[f_{i\to1}] + \mathsf{H}[f_{i\to-1}]\big).$$

If U_d bounds Δ_1/I_1 for symmetric functions in dimension d, then revealing coordinates one at a time gives $\mathsf{H}[f] \le n^{-1}(\sum_{d=1}^n U_d)\mathsf{I}[f]$ (Lemma 4.2). A single uniform bound cannot be smaller than $K = 12 - \frac{9}{2}\log_2 3$, because OR_3 has $\Delta_1/\mathsf{I}_1 = K$. Averaging over dimensions charges this obstruction only at dimension three. We obtain a stronger base by restricting several coordinates together through dimension 20, then add the dimension-specific one-step bounds. This seeded average proves Theorem A.

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Theorem B. (i) *For every $n \geq 1$ and every symmetric Boolean f on n variables, $\Delta_1[f] \leq K\,\mathsf{I}_1[f]$, with equality for* OR_3.

(ii) *In high dimension the step is cheaper:*

$$\limsup_{n\to\infty} \sup_f \frac{\Delta_1[f]}{\mathsf{I}_1[f]} \leq C_* < 4.637,$$

where the supremum is over nonconstant symmetric f on n variables and

$$C_* = \frac{1}{\ln 2}\Big[\frac{14}{5} + \frac{89}{50}\,e^{-10} I_0(10) + \frac{48}{23\sqrt{40\pi}} + \frac{e^{-z}}{z}\Big] = 4.6364236659\ldots, \qquad z = 40\Big(\frac{23}{48}\Big)^2.$$

Part (i) is computer-assisted. We check dimensions up to 15 exhaustively. We verify the scalar inequalities separately in dimensions 16 through 255 and on 23 blocks covering dimensions 256 through 8,388,608. The analytic estimate of Proposition 4.6, with parameter $a = 1/30$, bounds every later one-step ratio by 4.654017. Section 7 gives the certificates and the seeded ledger. Its largest upper bound is less than 4.657435, at dimension 449; this is a maximum of bounds, not a claimed value of H/I.

Both parts rest on a noise-stability estimate for derivatives, which is of independent interest. Write T_λ for the noise operator, $\mathrm{Stab}_\lambda[g] = \langle g, T_\lambda g\rangle = \sum_S \lambda^{|S|}\widehat{g}(S)^2$, and I_0 for the modified Bessel function of order zero.

Theorem C. *Let f be a symmetric Boolean function on $n \geq 2$ variables, $N = n-1$, and let $g = D_n f$ be its derivative in the last coordinate, a function on $\{-1,1\}^N$. For $0 \leq \lambda \leq 1$,*

$$\mathrm{Stab}_\lambda[g] \leq e^{-v} I_0(v)\,\|g\|_2^2, \qquad v = \frac{N(1-\lambda^2)}{4}.$$

Since $e^{-v}I_0(v) \sim (2\pi v)^{-1/2}$, this improves the derivative estimate used in [7] by a factor that tends to 2 when $\lambda \to 1$ and $v \to \infty$ together. The proof uses three facts: the derivative of a symmetric function is an alternating sum of indicators of Hamming slices; an alternating sum of selected values of a unimodal sequence is at most its largest value; and every atom of a sum of independent Bernoulli variables is bounded by Fourier inversion.

The restriction step loses most at the zero-level coefficient $\widehat{f}(\emptyset)$, which appears in Δ_1 but carries no influence. The asymptotic argument in part (ii) bounds its contribution separately by $O(n^{-1/2})$; that is too crude in moderate dimensions. We instead bound it by a concave function of the mass μ_0 that the derivative puts on level zero, and use a tangent of that function, so that the zero level enters the same linear program as the other levels (Lemma 4.3). This is what makes the finite range tractable.

The noise-stability estimate also gives a bound that needs no computer at all.

Theorem D. *Every symmetric Boolean function f satisfies*

$$\mathsf{H}[f] \leq \Big(\log_2(4+e) + 2 + \frac{1}{\ln 2}\Big)\mathsf{I}[f] < 6.191\,\mathsf{I}[f].$$

Here the entropy splits as $\mathsf{H} = L + B$, where L is the entropy of the level weights $W_k = \sum_{|S|=k}\widehat{f}(S)^2$ and $B = \sum_k W_k \log_2 \binom{n}{k}$ measures the spread inside the levels. The bound $L \leq \log_2(4+e)\mathsf{I}$ holds for every Boolean function; it is Lemma 2.1 of [8]. We prove $B < (2 + 1/\ln 2)\mathsf{I}$ for nonconstant symmetric f by integrating the envelope $e^{-v}I_0(v)$ against a harmonic-number kernel; a Frullani identity evaluates the main part exactly, and two inequalities for $e^{-x}I_0(x)$ and $e^{-x}I_1(x)$ control the rest.

The constant 4 is a natural lower bound.

Theorem E. *For the OR function on n variables (and for AND),*

$$\frac{\mathsf{H}[\mathrm{OR}_n]}{\mathsf{I}[\mathrm{OR}_n]} = 4 - \frac{4 - 2/\ln 2}{n} + O(2^{-n}).$$

Hence the optimal constant for symmetric functions lies in [4, 4.657435].

That OR approaches 4 is classical; we include the exact expression because AND and OR are the maximizers among all symmetric functions in every dimension we computed (Section 7).

Notation. Logarithms are natural unless written $\log_2$. Entropies are in bits. For a symmetric function we write $F(w) \in \{-1, 1\}$ for its value on weight w, $\widehat{f}_k$ for the common coefficient at level k, and $W_k = \binom{n}{k}\widehat{f}_k^2$. We put

$$\Phi(x) = e^{-x} I_0(x), \qquad G(x) = e^{-x} I_1(x) \qquad (x \geq 0),$$

so that $\Phi(x) = \frac{1}{\pi}\int_0^\pi e^{-x(1-\cos\theta)}\,d\theta$ and $\Phi' = G - \Phi$. Finally $\mathrm{E}_1(x) = \int_x^\infty e^{-u}u^{-1}\,du$ and $H_N = \sum_{i\leq N} 1/i$.

2. The derivative of a symmetric function

For $n \geq 2$ let $N = n - 1$ and $g = D_n f$, where $D_n f(x) = \frac{1}{2}(f(x,1) - f(x,-1))$ for $x \in \{-1,1\}^N$. Then $\widehat{g}(T) = \widehat{f}(T \cup \{n\})$, so g is symmetric with

$$W_j[g] = \binom{N}{j}\widehat{f}_{j+1}^2 = \frac{(j+1)W_{j+1}}{n}, \qquad \|g\|_2^2 = \sum_j W_j[g] = \mathsf{I}_n[f] = \frac{\mathsf{I}[f]}{n}. \tag{1}$$

Its values are $g(x) = \frac{1}{2}(F(w) - F(w+1))$ with $w = |x|$. They are nonzero exactly on the weights $w_0 < w_1 < \dots$ where F changes value, and there they alternate in sign. Up to a global sign,

$$g(x) = \sum_i (-1)^i\,\mathbf{1}[|x| = w_i]. \tag{2}$$

In particular $|g| = g^2$, so $\mathbb{E}|g| = \|g\|_2^2$. By symmetry the same holds for the derivative in any coordinate.

3. Noise stability of derivatives

Lemma 3.1. *Let $(a_y)_{y\geq 0}$ be a nonnegative unimodal sequence and $y_0 < y_1 < \cdots < y_m$ integers. Then $|\sum_i (-1)^i a_{y_i}| \leq \max_y a_y$.*

Proof. Write $a_y = \int_0^\infty \mathbf{1}[a_y > t]\,dt$. For each t the set $\{y : a_y > t\}$ is an interval, so the selected indices in it form a block of consecutive i, whose alternating sum is 0 or ± 1. The integrand vanishes for $t \geq \max a$. Integrate. □

Lemma 3.2. *Let Z be a sum of N independent Bernoulli variables with parameters in $\{p, 1-p\}$, and $v = Np(1-p)$. Then Z has a unimodal law and $\mathbb{P}[Z = y] \leq \Phi(v)$ for every integer y.*

Proof. Each Bernoulli law is log-concave and convolution preserves log-concavity of sequences supported on intervals, so the law of Z is log-concave and in particular unimodal. Each summand has characteristic function of modulus $\left(1 - 4p(1-p)\sin^2(\theta/2)\right)^{1/2}$. Using $1 - u \leq e^{-u}$,

$$|\mathbb{E}e^{i\theta Z}| \leq \exp\bigl(-2v\sin^2(\theta/2)\bigr) = e^{-v(1-\cos\theta)}.$$

Fourier inversion gives $\mathbb{P}[Z = y] \leq \frac{1}{2\pi}\int_{-\pi}^{\pi} |\mathbb{E}e^{i\theta Z}|\,d\theta \leq \Phi(v)$. □

Proof of Theorem C. Let $p = (1-\lambda)/2$. Given x with $|x| = r$, the weight of a λ-correlated copy y of x is $Z_r = \mathrm{Bin}(r, 1-p) + \mathrm{Bin}(N-r, p)$, of variance $Np(1-p) = v$. By (2),

$$\mathrm{Stab}_\lambda[g] = \mathbb{E}_x\Bigl[g(x)\sum_i (-1)^i\,\mathbb{P}[Z_{|x|} = w_i]\Bigr].$$

By Lemmas 3.1 and 3.2 the inner sum has absolute value at most $\Phi(v)$. Hence $\mathrm{Stab}_\lambda[g] \leq \Phi(v)\,\mathbb{E}|g| = \Phi(v)\|g\|_2^2$. □

Remark 3.3. For even N the atom bound before the step $1-u \le e^{-u}$ is attained at $r = y = N/2$. The constant $\sup_v \sqrt{v}\,\Phi(v) = 0.46882\ldots$ is the sharp uniform concentration constant for Bernoulli sums of Baillon, Cominetti and Vaisman [2]; we use the full function Φ rather than this constant.

4. The restriction step

Lemma 4.1. *Suppose $\Delta_1[f] \le K\,\mathsf{I}_1[f]$ for every symmetric f on at most m variables. Then $\mathsf{H}[f] \le K\,\mathsf{I}[f]$ for every symmetric f on at most m variables.*

Proof. Restrictions of a symmetric function are symmetric. For $i \ne 1$ the influence of i is the average of its influences in the two restrictions of x_1, so $\mathsf{I}[f] = \mathsf{I}_1[f] + \frac{1}{2}(\mathsf{I}[f_{1\to 1}] + \mathsf{I}[f_{1\to -1}])$. Induct on n, starting from constants. □

Lemma 4.2 (Averaging over dimensions)**.** *Suppose that $\Delta_1[g] \le U_d \mathsf{I}_1[g]$ for every symmetric Boolean function g on d variables. Then for every symmetric f on $n \ge 1$ variables,*

$$\mathsf{H}[f] \le \frac{1}{n}\sum_{d=1}^{n} U_d\,\mathsf{I}[f].$$

More generally, if $\mathsf{H}[g] \le C_m \mathsf{I}[g]$ in dimension m, then for $n \ge m$, $n \ge 1$,

$$\mathsf{H}[f] \le \frac{mC_m + \sum_{d=m+1}^{n} U_d}{n}\,\mathsf{I}[f]. \tag{3}$$

Let $\Delta_{[k]}[f]$ be the entropy lost after fixing k specified coordinates uniformly. If $\Delta_{[k]}[f] \le kB_{k,n}\mathsf{I}_1[f]$, one may choose the dimension bound

$$C_n = \frac{kB_{k,n} + (n-k)C_{n-k}}{n}. \tag{4}$$

One may minimize this choice over any available block sizes k.

Proof. A restriction of a symmetric weight table is a contiguous subtable, so it remains symmetric in the free coordinates. If i is free and Z fixes any set of other coordinates uniformly, then

$$\mathbb{E}_Z \mathsf{I}_i[f_Z] = \mathbb{E}_Z \mathbb{P}_{x_{\mathrm{free}}}[f_Z(x) \ne f_Z(x^{\oplus i})] = \mathbb{P}_x[f(x) \ne f(x^{\oplus i})] = \mathsf{I}_i[f].$$

This is conditional expectation of the indicator of a changing edge. It includes constant restrictions.

Reveal the coordinates in a fixed order and let F_d be the random function on the d unrevealed coordinates. For the next coordinate i_d,

$$\mathsf{H}[F_d] - \mathbb{E}[\mathsf{H}[F_{d-1}] \mid F_d] \le U_d \mathsf{I}_{i_d}[F_d].$$

On a branch with zero influence the symmetric function is constant, and both sides vanish. Taking expectations, influence preservation gives $\mathbb{E}\mathsf{I}_{i_d}[F_d] = \mathsf{I}_1[f] = \mathsf{I}[f]/n$. Summing over $d = 1, \ldots, n$ telescopes to $\mathsf{H}[f]$, since F_0 is constant. This proves the first assertion. For the second, stop at dimension m and use $\mathbb{E}\mathsf{H}[F_m] \le C_m \mathbb{E}\mathsf{I}[F_m] = mC_m\mathsf{I}[f]/n$. Finally, after fixing k coordinates, $\mathbb{E}\mathsf{I}[f_Z] = (n-k)\mathsf{I}[f]/n$. Add the block entropy loss to $\mathbb{E}\mathsf{H}[f_Z] \le C_{n-k}\mathbb{E}\mathsf{I}[f_Z]$ to obtain (4). Constants, including dimension zero, have $\mathsf{H} = \mathsf{I} = 0$ and need no division. No sign assumption on the entropy losses is used. □

For symmetric f on n variables, the restrictions to $x_1 = \pm 1$ have coefficients $\widehat{f}_s \pm \widehat{f}_{s+1}$ on the $\binom{N}{s}$ sets of size s in the remaining coordinates, $N = n-1$. With $\varphi(u) = u\log_2(1/u)$ this gives the exact formulas

$$\Delta_1 = \sum_{s=0}^{N} \binom{N}{s} \mathcal{D}(\widehat{f}_s, \widehat{f}_{s+1}), \qquad \mathsf{I}_1 = \sum_{s=0}^{N} \binom{N}{s} \widehat{f}_{s+1}^2, \tag{5}$$

$$\mathcal{D}(a,b) = \varphi(a^2) + \varphi(b^2) - \tfrac{1}{2}\big[\varphi((a+b)^2) + \varphi((a-b)^2)\big].$$

For OR_3 one finds $\Delta_1/\mathsf{I}_1 = 12 - \frac{9}{2}\log_2 3$. From here on we work in natural units, $\mathcal{D}_e = (\ln 2)\mathcal{D}$, and divide by $\ln 2$ at the end. Throughout, f is nonconstant, $g = D_1 f$ and

$$\mu_s = \binom{N}{s}\widehat{f}_{s+1}^{\,2}/\mathsf{I}_1 \qquad (0 \le s \le N),$$

a probability law on $\{0,\dots,N\}$ by (1).

4.1. **A tangent bound.** Let $h(t) = \mathcal{D}_e(\sqrt{t},1)$ for $t \ge 0$. For $t > 0$, $t \ne 1$, direct differentiation gives

$$h''(t) = \frac{1}{2t^{3/2}}\ln\Big|\frac{\sqrt{t}-1}{\sqrt{t}+1}\Big| < 0,$$

h' is continuous at $t = 1$ and tends to 0 from above, so h is increasing and concave and $(x,y) \mapsto \mathcal{D}_e(\sqrt{x},\sqrt{y}) = y\,h(x/y)$ is concave on the quadrant. Every tangent gives $\mathcal{D}_e(a,b) \le h'(t)a^2 + [h(t) - th'(t)]b^2$. Taking the tangent at $t_s = (n-s)/s$ for the term $s \ge 1$ of (5), and using $\binom{N}{s}/\binom{N}{s-1} = (n-s)/s$ to move the a^2 part to the neighbouring index, one obtains

$$\frac{(\ln 2)\Delta_1}{\mathsf{I}_1} \le L_0 + \mu_0\kappa(\tfrac{1}{n}) + \sum_{s=1}^{N}\mu_s\Big[\Psi(\tfrac{s}{n}) + \kappa(\tfrac{s+1}{n}) - \kappa(\tfrac{s}{n})\Big], \qquad L_0 = \frac{\mathcal{D}_e(\widehat{f}_0,\widehat{f}_1)}{\mathsf{I}_1}, \tag{6}$$

where $\Psi(x) = h(\frac{1-x}{x})$ and $\kappa(x) = \frac{1-x}{x}h'(\frac{1-x}{x})$, extended continuously by $\kappa(0) = 1$, $\kappa(1) = 0$. Both Ψ and $\kappa(\frac{s+1}{n}) - \kappa(\frac{s}{n})$ are explicit, so the right side of (6) is a linear function of μ plus the zero-level term L_0.

We also use $h(t) \le 3 + \ln(t + \frac{1}{6})$. With $p(y) = y^2\ln(y^2)$, $p(0) = 0$, the second-difference identity (valid since p'' is locally integrable) and Jensen's inequality for the triangular density $1 - |u|$ on $[-1,1]$, which has mean 0 and second moment $\frac{1}{6}$, give

$$h(t) = 3 + \int_{-1}^{1}(1-|u|)\ln\big((\sqrt{t}+u)^2\big)\,du \le 3 + \ln\big(t + \tfrac{1}{6}\big). \tag{7}$$

4.2. **Three facts about the derivative.**

(a) Let $m = |\widehat{f}_1| = |\mathbb{E}g|$. Since $|g| = g^2$, $m \le \mathsf{I}_1$. By (2) and Lemma 3.1 applied to the binomial weights, $m \le b_N := \max_j \binom{N}{j}2^{-N}$. By Parseval for f, $\widehat{f}_0^{\,2} + nm^2 \le 1$.

(b) Consecutive values of g on adjacent weights have nonpositive product, so $\|D_j g\|_2^2 = \frac{1}{2}(\|g\|_2^2 - \mathbb{E}[g(x)g(x^{\oplus j})]) \ge \frac{1}{2}\|g\|_2^2$ for each j, and summing over j gives $\mathbb{E}_\mu s \ge N/2$.

(c) By Theorem C, $\sum_s \mu_s\lambda^s \le \Phi\big(N(1-\lambda^2)/4\big)$ for $0 \le \lambda \le 1$. With $\lambda = e^{-t/n}$,

$$\mathbb{E}_\mu e^{-ts/n} \le \Phi\big(v_n(t)\big), \qquad v_n(t) = \frac{N(1-e^{-2t/n})}{4} \qquad (t > 0). \tag{8}$$

4.3. **The zero level.**

Lemma 4.3. *Let $C = n - \frac{1}{6}$ and $J_C(z) = z\big[3 + \ln(1 - Cz^2) - 2\ln z\big]$ for $0 < z < C^{-1/2}$. Then $L_0 \le J_C(\mu_0)$. The function J_C is concave, so for every $0 < u < C^{-1/2}$*

$$L_0 \le d_C(u) + j_C(u)\,\mu_0, \qquad d_C(u) = \frac{2u}{1-Cu^2}, \quad j_C(u) = J_C'(u) = 1 - 2\ln u + \ln(1-Cu^2) - \frac{2Cu^2}{1-Cu^2}.$$

Proof. If $m = 0$ then $L_0 = 0$ and $\mu_0 = 0$. Otherwise $\mu_0 = m^2/\mathsf{I}_1 \le m \le n^{-1/2}$ by (a). Since $\mathcal{D}_e(a,b) = b^2 h(a^2/b^2)$, h increases, and $\widehat{f}_0^{\,2} \le 1 - nm^2$, the bound (7) gives

$$L_0 = \mu_0\,h(\widehat{f}_0^{\,2}/m^2) \le \mu_0\big[3 + \ln(m^{-2} - C)\big] \le \mu_0\big[3 + \ln(\mu_0^{-2} - C)\big] = J_C(\mu_0),$$

using $m \geq \mu_0$ in the last step. Differentiating twice,

$$J_C''(z) = -\frac{2}{z} - \frac{6Cz}{1-Cz^2} - \frac{4C^2z^3}{(1-Cz^2)^2} < 0,$$

and the tangent at u has intercept $J_C(u) - uJ_C'(u) = d_C(u)$. □

Lemma 4.4 (The shift estimate)**.** *The function κ is increasing on $[0, \frac{1}{2}]$ and $\kappa(\frac{1}{2}) = 2\ln 2$. For $0 < x < 1$, $x \neq \frac{1}{2}$,*

$$\kappa'(x) \leq B(x) := 2 + 2\ln\frac{1}{|1-2x|}. \tag{9}$$

Consequently, whenever $0 \leq x \leq x + d \leq 1$,

$$\kappa(x+d) - \kappa(x) \leq d\big(4 + 2\ln(1/d)\big) \leq d\big(6 + 4\ln(1/d)\big).$$

Proof. For $t \geq 1$, expansion of the entropy pair gives

$$h(t) = \ln t + 3 - \sum_{j\geq 1} \frac{t^{-j}}{j(j+1)(2j+1)}.$$

For $0 < x < \frac{1}{2}$, put $r = x/(1-x)$. Then

$$\kappa(x) = 1 + \sum_{j\geq 1} \frac{r^j}{(j+1)(2j+1)},$$

which increases; at $r = 1$ the sum is $2\ln 2 - 1$. Termwise differentiation on compact subintervals, $j/[(j+1)(2j+1)] \leq 1/(2j)$ and $(1-x)^{-2} \leq 4$ give

$$\kappa'(x) \leq \frac{2[-\ln(1-r)]}{r} \leq 2[1 - \ln(1-r)] \leq 2 - 2\ln(1-2x).$$

The middle inequality follows from $(1-r)[-\ln(1-r)] \leq r$, by integrating $1/(1-t) \leq 1/(1-r)$ on $[0, r]$. The last uses $1 - r = (1-2x)/(1-x) \geq 1 - 2x$.

For $x > \frac{1}{2}$, put $t = (1-x)/x < 1$. The identity $h(t) = th(1/t)$ gives

$$\kappa = t(2 - \ln t) - \sum_{j\geq 1} \frac{t^{j+1}}{j(2j+1)}, \qquad \frac{d\kappa}{dt} = 1 - \ln t - \sum_{j\geq 1} \frac{(j+1)t^j}{j(2j+1)}.$$

Since $(j+1)/[j(2j+1)] = 1/(2j) + 1/[2j(2j+1)]$ and $\sum_{j\geq 1} 1/[2j(2j+1)] \leq \frac{1}{2}$, we have

$$\frac{d\kappa}{dt} \geq \tfrac{1}{2} - \ln t + \tfrac{1}{2}\ln(1-t) \geq \tfrac{1}{2}\ln(1-t).$$

Multiplying by $dt/dx = -x^{-2}$ reverses the inequality. Thus

$$\kappa'(x) \leq -\frac{\ln(1-t)}{2x^2} \leq -2\ln(1-t) \leq -2\ln(2x-1),$$

which is stronger than (9). The explicit entropy formula, or these series at their endpoints, shows that κ is absolutely continuous, with integrable logarithmic singularities in its derivative. On an interval of length d the integral of $\ln(1/|1-2y|)$ is largest when the interval is centred at $1/2$, where it equals $d(1 - \ln d)$. Integrating (9) proves the stated shift bounds. □

4.4. **The finite range.**

Proposition 4.5. *Let $n \ge 2$, $C = n - \frac{1}{6}$, $0 < u < C^{-1/2}$, and let $\alpha,\ \beta \ge 0$, $w_j \ge 0$, $t_j > 0$ $(1 \le j \le J)$ be real numbers with*

$$\kappa(\tfrac{1}{n}) + j_C(u) \le \alpha + \textstyle\sum_j w_j,$$
$$\Psi(\tfrac{s}{n}) + \kappa(\tfrac{s+1}{n}) - \kappa(\tfrac{s}{n}) \le \alpha - \beta\tfrac{s}{n} + \textstyle\sum_j w_j e^{-t_j s/n} \qquad (1 \le s \le N).$$

Then every nonconstant symmetric f on n variables satisfies

$$\frac{(\ln 2)\Delta_1}{\mathsf{I}_1} \le \alpha + d_C(u) - \beta\frac{N}{2n} + \sum_j w_j\, \Phi(v_n(t_j)).$$

Proof. Insert Lemma 4.3 into (6), bound each coefficient of μ_s by the hypotheses, and use $\mathbb{E}_\mu s \ge N/2$ from (b) and (8). □

Proposition 4.5 also certifies whole ranges of dimensions. Fix $256 \le L \le n \le U$ and use $C = L - \frac{1}{6}$, which only enlarges J_C, and $\kappa(\frac{1}{n}) \le \kappa(\frac{1}{L})$. We bound the entire concave function J_C before taking its tangent; no monotonicity of the tangent slope or intercept separately is required.

The sharper derivative estimate gives a separate shift bound on each rational cell $[l, r]$. Put $h_L = 1/L$ and

$$x_0 = \min\{r, \max\{l, (1 - h_L)/2\}\}, \qquad M_{[l,r]} = \mathcal{G}(x_0 + h_L - \tfrac{1}{2}) - \mathcal{G}(x_0 - \tfrac{1}{2}),$$

where $\mathcal{G}(z) = 4z - 2z\ln(2|z|)$ for $z \ne 0$, and $\mathcal{G}(0) = 0$. Indeed $\mathcal{G}'(y - 1/2) = B(y)$. For $x = s/n \in [l, r]$, Lemma 4.4 gives

$$\kappa(x + 1/n) - \kappa(x) \le \int_x^{x+1/n} B(y)\, dy \le \int_x^{x+h_L} B(y)\, dy \le M_{[l,r]}.$$

In the second integral we extend $B(y) = 2 - 2\ln|1 - 2y|$ as far as $1 + h_L$ if needed. It is nonnegative there for $L \ge 256$. No value of κ outside $[0, 1]$ is used. The integral over a fixed length h_L increases until its starting point is $(1 - h_L)/2$ and decreases afterwards: its derivative is $B(x + h_L) - B(x)$, whose sign follows by comparing the distances from $1/2$. This proves the formula for the maximum over a cell, including cells crossing $1/2$.

Both Ψ and $x \mapsto \alpha - \beta x + \sum_j w_j e^{-t_j x}$ decrease. Therefore the zero test at $1/L$ and the cell inequalities

$$\Psi(l) + M_{[l,r]} \le \alpha - \beta r + \sum_j w_j e^{-t_j r} \tag{10}$$

on a rational covering of $[1/U, 1]$ prove all the hypotheses. On the objective side, $N/(2n) \ge (L - 1)/(2L)$ and $v_n(t)$ increases in n. To see the latter, set $z = 2t/n$ and differentiate:

$$4\frac{d}{dn} v_n(t) = 1 - (1 + z)e^{-z} + \frac{z}{n} e^{-z} > 0.$$

Since Φ decreases, a certified block bound is

$$U_{[L,U]} = \frac{\alpha + d_{L-1/6}(u) - \beta(L - 1)/(2L) + \sum_j w_j \Phi(v_L(t_j))}{\ln 2}. \tag{11}$$

4.5. **High dimension.** For the analytic range we use the pointwise majorant

$$\Psi(x) \le U(x) := \mathrm{E}_1(40x) + \frac{553}{100} - \frac{273}{50}x + \frac{89}{50}e^{-20x} \qquad (0 < x \le 1). \tag{12}$$

It is verified by outward-rounded interval arithmetic on 11,900 cells covering $[10^{-4},1]$, using that Ψ and U are decreasing (Appendix A), with minimum certified slack above 0.02288; for $0<x\le 10^{-4}$ one uses $\Psi(x)\le 3-\ln x$, $\mathrm{E}_1(40x)\ge -\gamma-\ln(40x)$ and $e^{-20x}\ge 1-20x$. We also use, with $r_0=\frac{23}{48}$,

$$\Phi(v)\le \frac{24}{23\sqrt{2\pi v}}+e^{-2r_0^2 v},\qquad \bar J:=\frac{48}{23\sqrt{40\pi}}+\frac{e^{-z}}{z}\ge\int_{40}^{\infty}\Phi(t/2)\frac{dt}{t},\quad z=40r_0^2. \tag{13}$$

These follow from $\Phi(v)=\frac{2}{\pi}\int_0^{\pi/2}e^{-2v\sin^2 u}du$, $\sin u\ge 2r_0u$ on $[0,\frac12]$ and $\sin u\ge r_0$ on $[\frac12,\frac{\pi}{2}]$, and $\mathrm{E}_1(z)\le e^{-z}/z$.

Proposition 4.6. *Fix $0<a\le\frac15$. For $n\ge 40/a$ put $N=n-1$, $b=\sqrt{2/(\pi N)}$, $c=\frac{N}{n}\cdot\frac{1-e^{-2a}}{2a}$, $v_T=N(1-e^{-2a})/4$ and $A=-\frac12-\ln a+a$. Then every nonconstant symmetric f on n variables satisfies $(\ln 2)\Delta_1/\mathsf{I}_1\le R_a(n)$, where*

$$R_a(n)=b\big[3+\ln(1+b^{-2})+2\ln 2\big]+\frac{6+4\ln n}{n}+\frac{14}{5}+\frac{273}{100n}+\bar J \\ +\Phi(20c)\ln\frac1c+e^aA\Big[\frac{24}{23\sqrt{2\pi v_T}}+e^{-2r_0^2v_T}\Big]+\frac{89}{50}\Phi(v_n(20)).$$

Every term of R_a is nonincreasing for $n\ge 40/a$. In particular, $a=1/30$ is valid for $n\ge 1200$, and $R_{1/30}(8{,}388{,}608)/\ln 2<4.654016341712388$.

Proof. By (a) and (7) with $\widehat{f}_0^{\,2}\le 1$, $L_0\le m[3+\ln(1+m^{-2})]$, and $u\mapsto u[3+\ln(1+u^{-2})]$ increases. Also $\mu_0\le m$, $\kappa(\frac1n)\le\kappa(\frac12)=2\ln 2$, and $b_N\le b$ (for even N write $b_N=\frac{2}{\pi}\int_0^{\pi/2}\cos^N u\,du$ and use $\cos u\le e^{-u^2/2}$; for odd N, $b_N=b_{N+1}$). So the zero-level terms of (6) are at most the first term of R_a, and by Lemma 4.4 the shifts contribute at most $(6+4\ln n)/n$.

For $s\ge 1$ apply (12) at $x=s/n$. The linear part gives at most $\frac{553}{100}-\frac{273}{50}\mathbb{E}_\mu\frac{s}{n}\le\frac{14}{5}+\frac{273}{100n}$ by (b), and the last part at most $\frac{89}{50}\Phi(v_n(20))$ by (8). For the E_1 part let $P(t)=\sum_{s\ge1}\mu_s e^{-ts/n}$, so that $\sum_{s\ge1}\mu_s\,\mathrm{E}_1(40s/n)=\int_{40}^{\infty}P(t)\frac{dt}{t}$ by Tonelli. Put $T=an\ge 40$. For $40\le t\le T$, since $(1-e^{-2y})/(2y)$ decreases, $v_n(t)\ge ct/2$, and (8) gives $P(t)\le\Phi(ct/2)$; hence

$$\int_{40}^{T}P(t)\frac{dt}{t}\le\int_{40c}^{\infty}\Phi(y/2)\frac{dy}{y}\le\bar J+\Phi(20c)\ln\frac1c.$$

For $t\ge T$ and $s\ge1$, $e^{-ts/n}\le e^{-(t-T)/n}e^{-Ts/n}$, so $P(t)\le e^{-(t-T)/n}\Phi(v_T)$ and

$$\int_T^{\infty}P(t)\frac{dt}{t}\le\Phi(v_T)\int_a^{\infty}e^{a-y}\frac{dy}{y}=e^a\,\mathrm{E}_1(a)\Phi(v_T)\le e^aA\,\Phi(v_T),$$

using $\mathrm{E}_1(a)\le-\gamma-\ln a+a$ and $\gamma>H_6-\ln 7>\frac12$ (since $H_6=49/20$ and $\ln 7<39/20$). Bound $\Phi(v_T)$ by (13). Removing the zero level from P is what keeps this integral finite.

For the monotonicity, the first term equals $\sqrt{2/\pi}\,[3+2\ln 2+\ln(1+\pi N/2)]/\sqrt{N}$, whose derivative in N is negative; $(6+4\ln n)/n$ decreases for $n\ge1$; c increases in n and stays in $(0,1)$; and v_T and $v_n(20)$ increase in n while Φ decreases. Here $A>0$ is fixed, and $c=(1-1/n)(1-e^{-2a})/(2a)$ increases within $(0,1)$. Thus $\Phi(20c)\ln(1/c)$ is a product of nonnegative decreasing factors. The analytic upper bound for $\Phi(v_T)$ in (13) decreases as well. More explicitly, writing $z=\pi N/2$, twice the derivative of the zero-level term, multiplied by $N^{3/2}/\sqrt{2/\pi}$, is

$$\frac{2z}{1+z}-[3+2\ln 2+\ln(1+z)]<0.$$

The only lower threshold used in the integral split is $T=an\ge40$. The bound $\mathrm{E}_1(a)\le-\gamma-\ln a+a$ follows by integrating $(1-e^{-x})/x\le1$; hence the whole proof holds for every stated a. The value at $a=1/30$ and $n=8{,}388{,}608$ is certified in interval arithmetic. □

4.6. **Proofs of Theorems B and A.**

Proof of Theorem B(i). Dimensions $n \le 15$. The Fourier coefficients of a symmetric f are integers divided by 2^n, given by Krawtchouk transforms. For each of the $\sum_{n\le 15} 2^{n+1} = 131{,}068$ symmetric functions we compute Δ_1 and I_1 from (5), with every logarithm enclosed between rational bounds from the series of artanh with an explicit remainder. Equality $\Delta_1 = K\mathsf{I}_1$ holds for $n = 3$ at OR, AND and their negations, where the value is an exact combination of $\log_2 2$ and $\log_2 3$; for every other function the upper enclosure of Δ_1/I_1 is below K.

Dimensions $16 \le n \le 255$. For each n we apply Proposition 4.5 with $J = 2$ and decimal rational parameters, checking all n hypotheses in outward-rounded interval arithmetic at 65 digits. These are 32,520 inequalities in total.

Dimensions $256 \le n \le 8{,}388{,}608$. We use the block form of Proposition 4.5 and the cell inequalities (10) on 23 contiguous blocks, with 65,139 rational covering cells. The full table appears in Section 7.

Dimensions $n \ge 8{,}388{,}608$. Proposition 4.6 with $a = 1/30$.

The bounds obtained are listed in Section 7; each is below K. □

Proof of Theorem A. For $n \le 20$ we use the certified values $B_{k,n}$ for $2 \le k \le \min(5, n)$ described in Section 7. Starting from $C_0 = C_1 = 0$, define C_n as the minimum of the right sides of (4) over these k. This gives $C_n < 4$ for $n \le 20$ and $C_{20} < 3.944265467369$.

For $21 \le n \le D := 8{,}388{,}608$, use the one-step bounds U_d of Propositions 4.5 and 4.6, and put

$$S_{20} = 20C_{20}, \qquad S_n = S_{n-1} + U_n.$$

Lemma 4.2 gives $\mathsf{H}[f]/\mathsf{I}[f] \le S_n/n$. The exact rational ledger in Section 7 verifies

$$\max_{20\le n\le D} \frac{S_n}{n} = 4.657434471752612150384327653 8\ldots < c,$$

with its maximum at $n = 449$. This decimal denotes the recorded rational upper bound, not an extremal value of H/I. For $d > D$, Proposition 4.6 gives $U_d \le 4.654016341712388 < c$. If $S_n \le nc$ and $U_{n+1} \le c$, then $S_{n+1} \le (n+1)c$. This proves every remaining dimension. Constants have $\mathsf{H} = \mathsf{I} = 0$. □

Proof of Theorem B(ii). Take any sequence $n \to \infty$ with nonconstant symmetric f_n, and pass to a subsequence along which the law of s/n under μ converges weakly to a law ν on $[0, 1]$. By (b), $\mathbb{E}_\nu X \ge \frac{1}{2}$. Letting $n \to \infty$ in (8),

$$\mathbb{E}_\nu e^{-tX} \le \Phi(t/2) \qquad (t > 0), \tag{14}$$

and in particular $\nu(\{0\}) = 0$. In (6) the zero-level terms are at most $b_N[3 + \ln(1 + b_N^{-2}) + 2\ln 2] \to 0$, as in the proof of Proposition 4.6, and the shifts at most $(6 + 4\ln n)/n \to 0$. The function Ψ is unbounded at 0 but satisfies $\Psi(x) \le 3 + \ln(1/x)$, and (8) with $t = n/m$ gives $\mu[s \le m] \le e\,\Phi\big(N(1 - e^{-2/m})/4\big) \le C\sqrt{m/N}$ for an absolute C. Summing over dyadic ranges, the contribution of $1 \le s \le \delta n$ is $O(\sqrt{\delta}(1+\ln(1/\delta)))$ uniformly in n. This uniform integrability and weak convergence give $\sum_{s\ge 1} \mu_s \Psi(s/n) \to \int \Psi\, d\nu$. Hence $\limsup(\ln 2)\Delta_1/\mathsf{I}_1 \le \sup_\nu \int \Psi\, d\nu$, the supremum over laws on $(0, 1]$ with $\mathbb{E}_\nu X \ge \frac{1}{2}$ and (14). Since $\mathrm{E}_1(40x) = \int_{40}^\infty e^{-tx}\frac{dt}{t}$, the majorant (12), Tonelli, (14) and $\mathbb{E}_\nu X \ge \frac{1}{2}$ give

$$\int \Psi\, d\nu \le \int_{40}^\infty \Phi(t/2)\frac{dt}{t} + \frac{553}{100} - \frac{273}{100} + \frac{89}{50}\Phi(10) \le \bar{J} + \frac{14}{5} + \frac{89}{50}\Phi(10).$$

Dividing by $\ln 2$ gives C_*, and a positive series for $I_0(10)$ certifies $\Phi(10) < 0.127833337164$, so $C_* < 4.637$. □

5. A bound without computation

For symmetric f every set of size k carries the same coefficient, so

$$\mathsf{H}[f] = \sum_{k=0}^{n} \binom{n}{k} \widehat{f}_k^{\,2} \log_2 \frac{1}{\widehat{f}_k^{\,2}} = \underbrace{\sum_k W_k \log_2 \frac{1}{W_k}}_{L} + \underbrace{\sum_k W_k \log_2 \binom{n}{k}}_{B}. \tag{15}$$

The level weights satisfy $\sum_k W_k = 1$ and $\sum_k k W_k = \mathsf{I}[f]$.

Lemma 5.1 ([8, Lemma 2.1])**.** *Every Boolean function satisfies $L \le \log_2(4+e)\,\mathsf{I}[f]$.*

Proof. We may assume f is not constant. Let $\varepsilon = 1 - W_0 > 0$ and $Q(k) = W_k/\varepsilon$ for $k \ge 1$. Put $a = \log_2(1 + e/4)$, so that $(2^a - 1)2^{-ak}$, $k \ge 1$, is a probability law. Gibbs' inequality against this law gives $\mathsf{H}(Q) \le a\,\mathbb{E}_Q k - \log_2(2^a - 1)$. Since $L = h_2(\varepsilon) + \varepsilon \mathsf{H}(Q)$, $\varepsilon \mathbb{E}_Q k = \mathsf{I}$ and $h_2(\varepsilon) \le \varepsilon \log_2(1/\varepsilon) + \varepsilon \log_2 e$, we get

$$L \le a\mathsf{I} + \varepsilon\big[\log_2(1/\varepsilon) + \log_2 e - \log_2(e/4)\big] = a\mathsf{I} + \varepsilon\big[\log_2(1/\varepsilon) + 2\big].$$

Let $p = \min\{\mathbb{P}[f=1], \mathbb{P}[f=-1]\}$, so $\varepsilon = 4p(1-p)$. The edge-isoperimetric inequality [6, Chapter 2] gives $\mathsf{I} \ge 2p \log_2(1/p)$, and

$$2p \log_2 \frac{1}{p} - \varepsilon\Big(1 + \tfrac{1}{2}\log_2 \frac{1}{\varepsilon}\Big) = 2p\big[p \log_2 \tfrac{1}{p} + (1-p)\log_2(1-p)\big] \ge 0,$$

because the bracket is concave on $[0, \frac{1}{2}]$ (its second derivative is $(2p-1)/(p(1-p)\ln 2)$) and vanishes at both ends. Hence $\varepsilon[\log_2(1/\varepsilon) + 2] \le 2\mathsf{I}$ and $L \le (a+2)\mathsf{I} = \log_2(4+e)\mathsf{I}$. □

Lemma 5.2. *Let f be symmetric on $n \ge 2$ variables with $\mathsf{I}[f] > 0$, and $N = n - 1$. Then*

$$B \le \frac{1 + C_N}{\ln 2}\,\mathsf{I}[f], \qquad C_N = \int_0^1 \frac{\Phi\big(Nt(2-t)/4\big) - (1-t)^N}{t}\,dt.$$

Proof. Using $\binom{n}{k} \le (en/k)^k$,

$$B \ln 2 \le \sum_{k \ge 1} k W_k \Big(1 + \ln \frac{n}{k}\Big) = \mathsf{I}\Big(1 + \mathbb{E}_q \ln \frac{n}{j+1}\Big), \qquad q_j = \frac{(j+1)W_{j+1}}{\mathsf{I}},\ 0 \le j \le N.$$

By (1), $q_j = W_j[g]/\|g\|_2^2$. Comparing the sum with an integral, $\ln \frac{n}{j+1} \le H_N - H_j = \int_0^1 \frac{\rho^j - \rho^N}{1-\rho}\,d\rho$. Therefore

$$\mathbb{E}_q \ln \frac{n}{j+1} \le \int_0^1 \frac{s(\rho) - \rho^N}{1-\rho}\,d\rho, \qquad s(\rho) = \sum_j q_j \rho^j = \frac{\mathrm{Stab}_\rho[g]}{\|g\|_2^2}.$$

By Theorem C, $s(\rho) \le \Phi\big(N(1-\rho^2)/4\big)$. Substituting $\rho = 1 - t$, so that $1 - \rho^2 = t(2-t)$, gives the claim. The integrand of C_N is $O(1)$ near $t = 0$. □

We need three facts about Φ and G.

Lemma 5.3. (i) *$\sqrt{w}\,\Phi(w) > 1/\sqrt{2\pi}$ for every $w \ge \frac{1}{2}$.*

(ii) *$J(X) := \int_0^X u\,\big(-\Phi'(u)\big)\,du = XG(X)$ for every $X \ge 0$.*

(iii) *$G(X) < 1/\sqrt{2\pi X}$ for every $X > 0$.*

Proof. (i) Let $h(w) = \sqrt{w}\,\Phi(w)$. Substituting $I_0 = e^w \Phi$ into the Bessel equation $I_0'' + I_0'/w - I_0 = 0$ gives

$$h'' + 2h' + \frac{h}{4w^2} = 0.$$

At a stationary point $h'' = -h/(4w^2) < 0$, so h has no local minimum in $(0, \infty)$. At the left end $h(\frac{1}{2}) \ge e^{-1/2}/\sqrt{2} > 1/\sqrt{2\pi}$, since $I_0 \ge 1$ and $\pi > e$. As $w \to \infty$, the substitution $\theta = y/\sqrt{w}$ and

dominated convergence (with $1-\cos\theta \geq 2\theta^2/\pi^2$ on $[0,\pi]$) give $h(w) \to \frac{1}{\pi}\int_0^\infty e^{-y^2/2}dy = 1/\sqrt{2\pi}$. If $h(w_0) \leq 1/\sqrt{2\pi}$ for some $w_0 > \frac{1}{2}$, then h restricted to some $[\frac{1}{2}, W]$ with $W > w_0$ attains its minimum at an interior point: if $h(w_0) < 1/\sqrt{2\pi}$ choose W with $h(W) > h(w_0)$, and if equality holds take any $W > w_0$, since then $h \geq h(w_0)$ on $[\frac{1}{2}, \infty)$ by the first case. This interior minimum is a local minimum, a contradiction.

(ii) From $I_0' = I_1$ and $I_1' = I_0 - I_1/x$ we get $\Phi' = G - \Phi$ and $G' = \Phi - G - G/x$. Hence $(xG)' = x(\Phi - G) = -x\Phi'$, and $xG(x) \to 0$ as $x \to 0$.

(iii) Integrating $I_1(X) = \frac{1}{\pi}\int_0^\pi e^{X\cos\theta}\cos\theta\, d\theta$ by parts and substituting $u = 1-\cos\theta$,

$$G(X) = \frac{X}{\pi}\int_0^\pi e^{-X(1-\cos\theta)}\sin^2\theta\, d\theta = \frac{X}{\pi}\int_0^2 e^{-Xu}\sqrt{u(2-u)}\, du$$
$$< \frac{\sqrt{2}X}{\pi}\int_0^\infty e^{-Xu}\sqrt{u}\, du = \frac{1}{\sqrt{2\pi X}}. \qquad \square$$

Proposition 5.4. *$C_N < \ln 4$ for every integer $N \geq 1$. More precisely $C_1 \leq 1$, $C_2 \leq 59/64$, and for $N \geq 3$*

$$C_N < \ln 4 - \frac{2-\sqrt{2}}{\sqrt{\pi N}} + \frac{1}{2N} < \ln 4.$$

Proof. Split $C_N = T_1 + T_2 + T_3$ with

$$T_1 = \int_0^1 \frac{\Phi(\frac{Nt(2-t)}{4}) - \Phi(\frac{Nt}{2})}{t}dt, \quad T_2 = \int_0^1 \frac{\Phi(\frac{Nt}{2}) - e^{-Nt}}{t}dt, \quad T_3 = \int_0^1 \frac{e^{-Nt} - (1-t)^N}{t}dt.$$

The term T_2. Write $\Phi(x) = \mathbb{E}_\theta e^{-x(1-\cos\theta)}$ with θ uniform on $[0,\pi]$. For each θ, Frullani's integral gives $\int_0^\infty (e^{-Nt(1-\cos\theta)/2} - e^{-Nt})\frac{dt}{t} = \ln\frac{2}{1-\cos\theta}$; the integrand is nonnegative, so by Tonelli

$$\int_0^\infty \frac{\Phi(Nt/2) - e^{-Nt}}{t}dt = \mathbb{E}_\theta \ln\frac{1}{\sin^2(\theta/2)} = -\frac{4}{\pi}\int_0^{\pi/2}\ln\sin u\, du = \ln 4.$$

On $t \geq 1$ we have $Nt/2 \geq \frac{1}{2}$, so Lemma 5.3(i) gives $\int_1^\infty \Phi(Nt/2)\frac{dt}{t} \geq \int_1^\infty \frac{dt}{\sqrt{\pi N}\, t^{3/2}} = \frac{2}{\sqrt{\pi N}}$. Since $\int_1^\infty e^{-Nt}\frac{dt}{t} = \mathrm{E}_1(N)$,

$$T_2 \leq \ln 4 - \frac{2}{\sqrt{\pi N}} + \mathrm{E}_1(N).$$

The term T_1. The function Φ is completely monotone, so $-\Phi'$ is positive and decreasing. With $a = Nt(2-t)/4$ and $b = Nt/2$ we have $a \geq Nt/4$ and $b - a = Nt^2/4$, so $\Phi(a) - \Phi(b) \leq \frac{Nt^2}{4}(-\Phi'(\frac{Nt}{4}))$. By Lemma 5.3(ii) and (iii),

$$T_1 \leq \int_0^1 \frac{Nt}{4}\Big(-\Phi'(\tfrac{Nt}{4})\Big)dt = \frac{4}{N}J(\tfrac{N}{4}) = G(\tfrac{N}{4}) < \sqrt{\frac{2}{\pi N}}.$$

The term T_3. Since $\int_0^1 \frac{1-(1-t)^N}{t}dt = H_N$ and $\int_0^1 \frac{1-e^{-Nt}}{t}dt = \ln N + \gamma + \mathrm{E}_1(N)$,

$$T_3 = H_N - \ln N - \gamma - \mathrm{E}_1(N) \quad (\geq 0).$$

Adding, the $\mathrm{E}_1(N)$ terms cancel and $H_N - \ln N - \gamma < 1/(2N)$ gives

$$C_N < \ln 4 + \frac{\sqrt{2}}{\sqrt{\pi N}} - \frac{2}{\sqrt{\pi N}} + \frac{1}{2N}.$$

The last two terms together are negative when $N > \pi/(4(2-\sqrt{2})^2)$, and this threshold is less than 3 because $\pi < 4 < 12(2-\sqrt{2})^2$, the last inequality following from $\sqrt{2} < 17/12$.

For $N = 1$, $\Phi \le 1$ gives $C_1 \le \int_0^1 \frac{1-(1-t)}{t} dt = 1$. For $N = 2$, $e^{-z} \le 1 - z + z^2/2$ together with $\mathbb{E}(1 - \cos\theta) = 1$ and $\mathbb{E}(1 - \cos\theta)^2 = \frac{3}{2}$ gives $\Phi(v) \le 1 - v + \frac{3}{4}v^2$. With $v = (1 - \rho^2)/2$,

$$C_2 \le \int_0^1 \frac{\frac{1-\rho^2}{2} + \frac{3}{16}(1-\rho^2)^2}{1-\rho}\, d\rho = \frac{3}{4} + \frac{3}{16} \cdot \frac{11}{12} = \frac{59}{64}.$$

Both are below $\ln 4$. □

Proof of Theorem D. For $n = 1$ every Boolean function is constant or $\pm x_1$ and $\mathsf{H} = 0$. For $n \ge 2$ and $\mathsf{I} > 0$, Lemma 5.2 and Proposition 5.4 give $B < \frac{1+\ln 4}{\ln 2}\mathsf{I} = (2 + \frac{1}{\ln 2})\mathsf{I}$, and Lemma 5.1 gives $L \le \log_2(4 + e)\mathsf{I}$. Add them using (15). □

Remark 5.5. The two parts are not tight together. For majority on $n = 2m + 1$ variables, $B/\mathsf{I} \to 2/\ln 2$, so the within-level constant cannot go below 2.885; and $L/\mathsf{I} \le 2.5176$ for every symmetric function with $n \le 16$, attained by AND and OR at $n = 6$ (Section 7). Numerically C_N increases to $\ln 4$: $C_1 = 0.6638$, $C_2 = 0.8911$, $C_3 = 0.9991$, $C_{10^6} = 1.3855$.

The majority limit in the remark is proved as follows. Write $n = 2m + 1$. The derivative of majority is the indicator of the middle slice of the $2m$-cube, and its normalized spectral weights are

$$q_{2j} = \binom{2j}{j}\binom{2m-2j}{m-j} 4^{-m}, \qquad q_{2j+1} = 0.$$

Thus j has the beta-binomial law with parameters $(m, \frac{1}{2}, \frac{1}{2})$, and j/m converges in law to the arcsine law on $(0, 1)$. By (1), B/I is the expectation of $\log_2 \binom{2m+1}{2j+1}/(2j + 1)$. Stirling's formula gives the interior limit $h(x)/(x \ln 2)$ with $h(x) = -x \ln x - (1 - x)\ln(1 - x)$. The estimates $q_{2j} \le C/\sqrt{(j + 1)(m - j + 1)}$ and $\ln \binom{n}{k}/k \le 1 + \ln(n/k)$ bound the contribution of $j \le \delta m$ by $C\sqrt{\delta}(1 + \ln(1/\delta)) + O(\ln(m)/\sqrt{m})$, and the other end contributes less, so the expectations converge. With $x = \sin^2 u$,

$$\mathbb{E}\frac{h(X)}{X} = \frac{2}{\pi}\int_0^{\pi/2} \left[-\ln(\sin^2 u) - \cot^2 u \ln(\cos^2 u)\right] du = 2,$$

by $\int_0^{\pi/2} \ln\sin u\, du = -\frac{\pi}{2}\ln 2$ and $\int_0^{\pi/2} \cot^2 u \ln(\cos^2 u)\, du = \pi(\ln 2 - 1)$. Hence $B/\mathsf{I} \to 2/\ln 2$.

6. AND and OR

Proof of Theorem E. Let f take the value -1 at exactly one point; this is OR or AND up to the signs of the inputs and the output, which change neither H nor I. Then $\widehat{f}(\emptyset) = 1 - 2^{1-n}$ and $\widehat{f}(S)^2 = 4^{1-n}$ for $S \ne \emptyset$. Hence, with $W_0 = (1 - 2^{1-n})^2$,

$$\mathsf{H}[f] = W_0 \log_2 \frac{1}{W_0} + (2^n - 1)\, 4^{1-n} \cdot 2(n - 1), \qquad \mathsf{I}[f] = n\, 2^{1-n}.$$

The second term divided by I is $(2 - 2^{1-n})\frac{2(n-1)}{n} = 4 - \frac{4}{n} + O(2^{-n})$, and $W_0 \log_2(1/W_0) = \frac{2^{2-n}}{\ln 2}(1 + O(2^{-n}))$ contributes $\frac{2}{n \ln 2} + O(2^{-n})$. □

At $n = 1024$ the ratio is $3.998911513751\ldots$.

7. Computations

All entropy ratios below use bits. The one-step bounds U_d bound Δ_1/I_1. In dimensions $n \le 15$ exact enumeration gives $U_d \le K$, with equality only in dimension 3. In dimensions $16 \le n \le 255$, Proposition 4.5 gives $U_d \le 4.848905$, with the largest certified bound at $n = 25$. The new block certificates give the following upper bounds, rounded upwards:

dimensions	one-step bound	cells
$256 \le n \le 294$	4.691397764337803	989
$295 \le n \le 339$	4.680231363778156	1,316
$340 \le n \le 390$	4.669771627054690	1,774
$391 \le n \le 449$	4.660834262012443	1,598
$450 \le n \le 517$	4.653071439774263	1,433
$518 \le n \le 595$	4.646337217003664	1,323
$596 \le n \le 685$	4.640615237927098	1,242
$686 \le n \le 788$	4.635610258129181	1,354
$789 \le n \le 907$	4.631114929966981	1,376
$908 \le n \le 1{,}044$	4.627616624048243	1,311
$1{,}045 \le n \le 1{,}201$	4.625229653517468	1,284
$1{,}202 \le n \le 1{,}382$	4.622146309801676	2,648
$1{,}383 \le n \le 1{,}590$	4.618720507476198	2,798
$1{,}591 \le n \le 1{,}829$	4.615719288500242	2,558
$1{,}830 \le n \le 2{,}104$	4.613032839769942	2,397
$2{,}105 \le n \le 2{,}420$	4.610575912037690	2,299
$2{,}421 \le n \le 2{,}499$	4.607196989753158	2,232
$2{,}500 \le n \le 9{,}999$	4.624103643056020	2,294
$10{,}000 \le n \le 16{,}384$	4.597828415472139	3,261
$16{,}385 \le n \le 65{,}535$	4.599485777517825	4,450
$65{,}536 \le n \le 262{,}143$	4.592695956170183	6,162
$262{,}144 \le n \le 1{,}048{,}575$	4.589502673515857	8,511
$1{,}048{,}576 \le n \le 8{,}388{,}608$	4.588487126623142	10,529

The blocks are contiguous and have 65,139 rational cells in total. At and above $D = 8{,}388{,}608$, the analytic bound is $U_d \le 4.654016341712388$. The multipliers and tangent points are exact decimal rationals. Each proposed block intercept is increased by $1/1000$ before certification. The optimizer supplies proposals only; its output is not a proof. We check the zero coefficient, the cell inequalities (10), exact coverage of $[1/U, 1]$, and the objective (11) by outward intervals at 65 decimal digits. For $\Phi(v)$ we use its positive Bessel series with a geometric remainder when $v \le 100$, and the upper bound (13) otherwise. A second implementation checks all 32,520 finite coefficient inequalities and all cells at 90 digits, using different entropy formulas and a different primitive for the shift integral.

The finite block base. For $2 \le k \le 5$ and $k \le n \le 20$, let $B_{k,n}$ be the largest value of $\Delta_{[k]}/(k\mathsf{I}_1)$ over nonconstant symmetric functions. We enumerate every weight table up to output complementation. If $A_s = 2^n \widehat{f}_s$ and $E_n = 4^n \mathsf{H}[f]$, then

$$E_n = 2n4^n - 2\sum_s \binom{n}{s} A_s^2 \log_2 |A_s|,$$

with zero summands omitted. Fixing k coordinates, r of weight one, leaves the subtable $F(r), \dots, F(r+n-k)$; write $E_{n-k,r}$ for its scaled entropy. Then

$$4^n \Delta_{[k]} = E_n - 2^k \sum_{r=0}^{k} \binom{k}{r} E_{n-k,r}, \qquad 4^n k\mathsf{I}_1 = k2^{n+1} \sum_{w:F(w)\neq F(w+1)} \binom{n-1}{w}.$$

Integer Krawtchouk transforms give the A_s. Parseval and the independent edge formula for influence are checked at every table. Rational logarithm enclosures come from

$$\ln u = 2\sum_{j=0}^{29} \frac{z^{2j+1}}{2j+1} + R, \qquad z = \frac{u-1}{u+1}, \quad 1 \le u < 2, \quad 0 \le R \le \frac{9}{4 \cdot 61 \cdot 3^{61}}.$$

Range reduction and outward integer rounding give fixed-point enclosures with denominator 2^{56}. A second complete pass strictly excludes every nonmaximizer; exact prime-log expressions prove all surviving ties. A separate integer and 144-bit logarithm audit checks every surviving maximum. The census covers 2,097,150 tables up to complementation through dimension 20, including constants, and supplies all 70 pairs (k, n). The recurrence (4) gives

$$C_{20} = \frac{92232807018024312555229592687247931457194510016451}{23384026197294446691258957323460528314494920687616} < 3.944265467369.$$

This is a certified upper bound. It is not identified with the exact maximum of H/I in dimension 20.

The averaged ledger. Seed the sum with $20C_{20}$ and add the U_d for $21 \le d \le D$. Exact rational addition gives

$$\max_{20 \le n \le D} \frac{20C_{20} + \sum_{d=21}^{n} U_d}{n}$$
$$= \frac{37308041988480180384659602316674779040565936050307820821254937 9}{80104276753121998324396300035352914529481654272000000000000000}.$$

This equals $4.65743447175261215038432765 38\ldots$ and the maximum occurs at $n = 449$. The claimed $c = 931487/200000$ is an upward rounding. We also sum by segment lengths and recompute the binding value and cutoff interface with 85-digit decimal arithmetic. On a segment with constant bound U, the ratio has the form $U + A/n$, so its maximum lies at an endpoint. This gives a further check of the finite maximum. At D the ledger is below 4.588797, and every later one-step bound is below c by Proposition 4.6. The seed $20C_{20}$ is essential to the description of this computation: the displayed ledger is not the pure sum $\sum_{d=1}^{n} U_d$.

The certificate scripts, rational multipliers, cells and ledger are included with the data. The computation guide gives the command for each statement and distinguishes the exhaustive certificates from numerical checks.

Over all symmetric functions with $n \le 16$ (exact integer Fourier coefficients, floating-point logarithms) the repository's check script finds:

quantity	maximum over symmetric f, $n \le 16$
H/I	3.93027, at AND/OR with $n = 16$
L/I	2.51755, at AND/OR with $n = 6$
B/I	1.80986
Δ_1/I_1	4.86767, at OR_3

In every dimension $2 \le n \le 16$ the maximum of H/I is attained by AND and OR. The script also checks the integral C_N by adaptive quadrature for $N \le 200$ and for N up to 10^6, Lemma 5.3 on dense grids, Theorem C on random symmetric derivatives with $n = 20, 40, 80$, the closed form of Theorem E, and the constants K and C_* at 40 digits. These checks support but are not part of the proofs; the computer-assisted parts use the exact enumeration for $n \le 15$, the block census through dimension 20, the interval certificates for Propositions 4.5 and 4.6, the exact rational ledger, and the certificate for (12).

8. Questions

Question 8.1. Is the optimal constant for symmetric functions equal to 4?

A single uniform restriction bound cannot be smaller than K, because of OR_3. Lemma 4.2 shows that this does not obstruct a smaller entropy–influence constant: one can retain the dimension-specific cost throughout the restriction chain.

For general Boolean functions, K does not bound the entropy loss at a suitable coordinate. In the $\{0,1\}$ convention, consider the majority of three disjoint AND pairs,

$$f = \mathrm{Maj}_3(x_1 \wedge x_2,\ x_3 \wedge x_4,\ x_5 \wedge x_6).$$

After converting the output to $\{-1,1\}$, every coordinate has

$$\frac{\Delta_i[f]}{\mathsf{I}_i[f]} = 5.1948716233791414624\ldots > K.$$

The OR of eight disjoint AND pairs on sixteen variables gives, at every coordinate,

$$\frac{\Delta_i[f]}{\mathsf{I}_i[f]} = 6.0037460565590882514\ldots > 6.$$

Both functions are transitive on their coordinates. Exact integer Walsh transforms and rational logarithm enclosures certify the two inequalities in the data.

Question 8.2. Is there an absolute constant C such that every nonconstant Boolean function has a coordinate i with $\mathsf{I}_i[f] > 0$ and $\Delta_i[f] \le C\,\mathsf{I}_i[f]$?

A positive answer would prove the entropy–influence conjecture with constant C by coordinate restriction and induction. The examples above show that such a C must exceed 6.

Appendix A. The scalar certificate for the majorant

The interval certificate for (12) uses two elementary lower bounds for E_1. For $0 < z \le 1$,

$$\mathrm{E}_1(z) \ge -0.578 - \ln z + \sum_{k=1}^{8} \frac{(-1)^{k+1} z^k}{k\,k!},$$

since an even truncation of the alternating series is a lower bound and $\gamma < H_{10} - \ln\frac{21}{2} < 0.578$. The last bound holds because $H_n - \ln(n+\frac{1}{2})$ decreases to γ: the integral of the convex function $1/x$ over $[n+\frac{1}{2}, n+\frac{3}{2}]$ exceeds its value $1/(n+1)$ at the midpoint. For $z > 1$,

$$\mathrm{E}_1(z) \ge e^{-z}\,\frac{z^2+8z+11}{z^3+9z^2+18z+6}.$$

Indeed $\mathrm{E}_1(z) = e^{-z}\,\mathbb{E}[(z+Y)^{-1}]$ for Y exponential with mean one, and for every quadratic polynomial P we have $(z+y)^{-1} \ge 2P(y) - (z+y)P(y)^2$, the difference being $[1-(z+y)P(y)]^2/(z+y)$. In the orthonormal basis $1,\ 1-y,\ 1-2y+y^2/2$ the compression of multiplication by $z+y$ to quadratic polynomials is

$$M = \begin{pmatrix} z+1 & -1 & 0 \\ -1 & z+3 & -2 \\ 0 & -2 & z+5 \end{pmatrix},$$

and optimizing over P gives $(M^{-1})_{00}$, which is the displayed rational function. The entries of M follow from $\mathbb{E}Y^k = k!$.

Both Ψ and U decrease: h increases, and $U'(x) = -e^{-40x}/x - 5.46 - 35.6\,e^{-20x} < 0$. So an outward interval lower bound for $U(r) - \Psi(l)$ proves (12) on a whole cell $[l,r]$. For $0 < x \le 10^{-4}$ the bounds in the text give

$$U(x) - \Psi(x) \ge 5.53 + 1.78 - 0.578 - \ln 40 - 3 - (5.46 + 20\cdot 1.78)\,10^{-4} > 0.$$

The value $\Phi(10)$ is enclosed using $I_0(10) = \sum_{k\geq 0} 25^k/(k!)^2$, with the terms through $k = 60$ summed and the rest bounded by a geometric series of ratio at most $25/62^2$. The resulting bound is $C_* < 4.636423665924$.

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [9] and working together inside the Brain framework [11], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [10], the Math Brain repository [9] and the Brain framework repository [11].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [11] and Math Brain [9]. In November 2026, Claude Opus 5.5 (Anthropic) and a team of Codex agents running OpenAI's GPT-6.1, all connected to the Brain, retrieved the spectral-level entropy lemma and related methods from the author's earlier work through Math Brain, derived the noise-stability estimate and the split of the entropy into level and within-level parts, proved the Bessel-function inequalities and the asymptotic restriction estimate, found the bound for the zero-level coefficient that closes the restriction step in every dimension, derived the averaging argument over successive dimensions and the sharper shift estimate, wrote and ran the interval certificates and the exhaustive computation for small dimensions, and searched the literature. Separate agents checked the averaging argument, the sharper shift estimate and the changed analytic cutoff, and recomputed the certificates with independently written code. Claude Opus 5.5 coordinated the agents, assembled the proofs, prepared the manuscript, and audited it, including the proofs, the numerical constants and the finite computations. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

`https://github.com/yspennstate/symmetric-fei-constant`

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [9].

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